1. Blackouts

1.1. Relevance to the driving task

For the purposes of this standard, the term 'blackout' means a transient impairment or loss of consciousness. Loss of consciousness is clearly incompatible with safe driving. The evidence for crash risk associated with various causes of blackout is discussed in the relevant chapters. This chapter provides guidance regarding the general management of blackouts, with cross-reference to relevant chapters as per Figure 6. Management of blackouts and driving.

1.2. General assessment and management guidelines

1.2.1. General considerations[1–4]

Blackouts may occur due to a range of mechanisms including:

- vasovagal syncope or 'faint', which accounts for more than 50 per cent of blackouts and may be due to factors such as hot weather, emotion or venepuncture but may also be due to more serious causes that may recur
- syncope due to other cardiovascular causes such as structural heart disease, arrhythmias or vascular disease
- epileptic seizure, which accounts for less than 10 per cent of blackouts
- other causes including metabolic causes (e.g. hypoglycaemia), drug intoxication or a sleep disorder.

Determination of the mechanism of a blackout may be straightforward based on history, investigations and specialist referral, and the person may be managed as per the appropriate chapter. Alternatively, it may require extensive cardiovascular and neurological investigations and referral to several specialists. People should be advised not to drive until the mechanism is ascertained and the corresponding standard met.

Some drivers may attribute a crash or driving mishap to a 'blackout' to excuse an event that occurred for some other reason such as inattention or distraction (e.g. a mobile phone conversation). There will also be a small proportion of cases in which a clear cause cannot be established.

1.2.2. Vasovagal syncope[5]

The most common cause of transient loss of consciousness is vasovagal syncope ('fainting'). Where this has been triggered by a well-defined provoking factor or a situation that is unlikely to recur while driving (e.g. prolonged standing, venepuncture or emotional situation), it is not necessary to restrict driving. However, vasovagal syncope may also result from other causes that are not so benign. In such cases, fitness to drive should be assessed according to the cardiovascular conditions standards for syncope (refer to section 2. Cardiovascular conditions).

1.2.3. Blackouts due to medical causes not covered in the standards

If the cause of the blackout is determined to be a medical condition not covered in the standards, then first principles regarding fitness to drive should be applied (refer to Part A section 2. Assessing fitness to drive – general guidance). Considerations include the likelihood of recurrence of blackout and the treatability of the condition as well as the nature of the driving task. There should also be an appropriate review period. A more stringent approach should be considered for commercial vehicle drivers.

1.2.4. Blackouts of undetermined mechanism

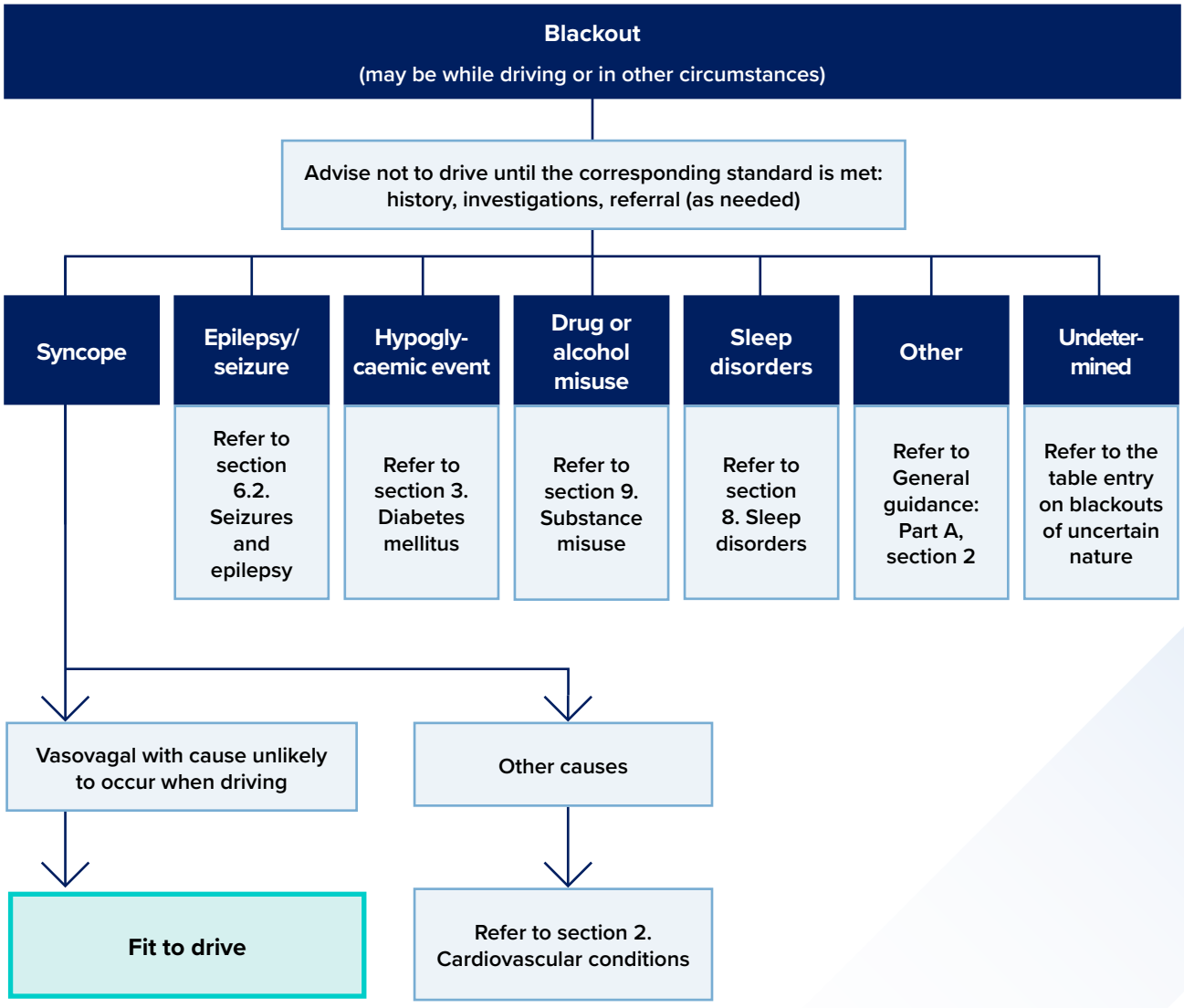
If, despite extensive investigation, the mechanism of a blackout cannot be determined, fitness to drive should be assessed according to section 1.3. Medical standards for licensing. The standards for blackout of undetermined mechanism are similar to those for seizure.

1.3. Medical standards for licensing

Requirements for unconditional and conditional licences are outlined in the following tables. Health professionals should familiarise themselves with the information in this chapter and the tabulated standards before assessing a person's fitness to drive.

Where a firm diagnosis has been made, the standard appropriate to the condition should be referred to in this publication (refer to Figure 6. Management of blackouts and driving). For blackouts due to medical causes not covered in the standard, refer to first principles (refer to Part A section 2. Assessing fitness to drive – general guidance). For blackouts where, after investigation, it is not possible to diagnose one of the conditions covered elsewhere in this publication, refer to the table for blackouts of uncertain nature over the page.

Figure 6. Management of blackouts and driving



Medical standards for licensing – blackouts of uncertain nature Health professionals should familiarise themselves with the information in this chapter and the tabulated standards before assessing a person's fitness to drive.		
Condition	**Private standards** (Drivers of cars, light rigid vehicles or motorcycles unless carrying public passengers or requiring a dangerous goods driver licence – refer to definition in Table 3)	**Commercial standards** (Drivers of heavy vehicles, public passenger vehicles or requiring a dangerous goods driver licence – refer to definition in Table 3)
Blackouts – episode(s) or impaired consciousness – of uncertain nature	**A person should not drive for 6 months following a single blackout of undetermined nature.** **A person should not drive for 12 months following two or more blackouts of undetermined nature separated by a 24-hour period.** A person is **not** fit to hold an **unconditional licence**: • if the person has experienced blackouts that cannot be diagnosed as syncope, seizure or another condition. If there has been a **single blackout** or more than one blackout within a 24-hour period, a **conditional licence** may be considered by the driver licensing authority subject to at least **annual review**, taking into account information provided by the **treating doctor** as to whether the following criterion is met: • there have been no further blackouts for at least **6 months**. If there have been **two or more blackouts** separated by at least 24 hours, a **conditional licence** may be considered by the driver licensing authority subject to at least **annual review**, taking into account information provided by the **treating doctor** as to whether the following criterion is met: • there have been no further blackouts for at least **12 months**.	**A person should not drive for 5 years following a single blackout of undetermined nature.** **A person should not drive for 10 years following two or more blackouts of undetermined nature separated by a 24-hour period.** A person is **not** fit to hold an **unconditional licence**: • if the person has experienced blackouts that cannot be diagnosed as syncope, seizure or another condition. If there has been a **single blackout** or more than one blackout within a 24-hour period, a **conditional licence** may be considered by the driver licensing authority subject to at least **annual review**, taking into account information provided by an **appropriate specialist** as to whether the following criterion is met: • there have been no further blackouts for at least **5 years**. If there have been **two or more blackouts** separated by at least 24 hours, a **conditional licence** may be considered by the driver licensing authority subject to at least **annual review**, taking into account information provided by an **appropriate specialist** as to whether the following criterion is met: • there have been no further blackouts for at least **10 years**.

Medical standards for licensing – blackouts of uncertain nature		
Condition	**Private standards** (Drivers of cars, light rigid vehicles or motorcycles unless carrying public passengers or requiring a dangerous goods driver licence – refer to definition in Table 3)	**Commercial standards** (Drivers of heavy vehicles, public passenger vehicles or requiring a dangerous goods driver licence – refer to definition in Table 3)
Exceptional cases	Where a person with one or more blackouts of undetermined mechanism does not meet the standards above for a conditional licence but may, in the opinion of the treating specialist, be safe to drive, a **conditional licence** may be considered by the driver licensing authority, subject to at least **annual review**: • if the driver licensing authority, after considering information provided by the **treating specialist(s)**, considers that the risk of a crash caused by a blackout is acceptably low.	Where a person with one or more blackouts of undetermined mechanism does not meet the standards above for a **conditional licence** but may, in the opinion of the treating specialist, be safe to drive, a **conditional licence** may be considered by the driver licensing authority, subject to at least **annual review**: • if the driver licensing authority, after considering information provided by the **treating specialist(s)**, considers that the risk of a crash caused by a blackout is acceptably low.

IMPORTANT: The medical standards and management guidelines contained in this chapter should be read in conjunction with the general information contained in Part A of this publication. Practitioners should give consideration to the following:

Licensing responsibility

The responsibility for issuing, renewing, suspending or cancelling a person's driver licence (including a conditional licence) lies ultimately with the driver licensing authority.

Licensing decisions are based on a full consideration of relevant factors relating to health and driving performance.

Conditional licences

For a conditional licence to be issued, the health professional must provide to the driver licensing authority details of the medical criteria not met, evidence of the medical criteria met, as well as the proposed conditions and monitoring requirements.

The presence of other medical conditions

While a person may meet individual disease criteria, concurrent medical conditions may combine to affect fitness to drive – for example, hearing, visual or cognitive impairment (refer to Part A section 2.2.7. Older drivers and age-related changes and section 2.2.8. Multiple medical conditions).

The nature of the driving task

The driver licensing authority will take into consideration the nature of the driving task as well as the medical condition, particularly when granting a conditional licence. For example, the licence status of a farmer requiring a commercial vehicle licence for the occasional use of a heavy vehicle may be quite different from that of an interstate multiple combination vehicle driver. The examining health professional should bear this in mind when examining a person and when providing advice to the driver licensing authority.

Reporting responsibilities

Patients should be made aware of the effects of their condition on driving and should be advised of their legal obligation to notify the driver licensing authority where driving is likely to be affected. The health professional may themselves advise the driver licensing authority as the situation requires (refer to section 3.3 and step 6).

Event/procedure	Minimum non-driving period (advisory) – private vehicle drivers	Minimum non-driving period (advisory) – commercial vehicle drivers
Other		
Deep vein thrombosis	2 weeks	2 weeks
Heart/lung transplant	6 weeks	3 months
Ventricular assist device	3 months	Not applicable
Pulmonary embolism	6 weeks	6 weeks
Syncope (due to cardiovascular causes)	4 weeks	3 months

2.2.3. High blood pressure

The cut-off blood pressure values at which a person is considered unfit to hold an unconditional licence do not reflect usual goals for managing hypertension. Rather, they reflect levels that are likely to be associated with sudden incapacity due to neurological events (e.g. stroke). The cut-off points are based on expert opinion.

It is a general requirement that conditional licences for commercial vehicle drivers are issued by the driver licensing authority based on the advice of an appropriate medical specialist and that these drivers are reviewed periodically by the specialist to determine their ongoing fitness to drive (refer to Part A section 4.4. Conditional licences). In the case of high blood pressure, ongoing fitness to drive may be assessed by the treating general practitioner, provided this is mutually agreed by the specialist and the general practitioner. The initial recommendation of a conditional licence must, however, be based on the opinion of the specialist.

2.2.4. Cardiac surgery (open chest)

Cardiac surgery may be performed for various reasons including valve replacement, excision of atrial myxoma and correction of septal defects. In some cases this is curative of the underlying disorder and so will not affect licence status for private or commercial vehicle drivers (refer also to Table 5. Suggested non-driving periods after cardiovascular events or procedures). In other cases, the condition may not be stabilised, and the effect on driving safety and hence on licence status needs to be individually assessed. All cardiac surgery patients should be advised regarding safety of driving in the short term as for any other post-surgery patient (e.g. taking into account the limitation of chest and shoulder movements after sternotomy).

2.2.5. Disorders of rate, rhythm or conduction

People with recurrent arrhythmias causing syncope or pre-syncope are usually not fit to drive. A conditional licence may be considered after appropriate treatment and an event-free non-driving period (refer to Table 5. Suggested non-driving periods after cardiovascular events or procedures).

Medical standards for licensing – cardiovascular conditions		
Condition	**Private standards** (Drivers of cars, light rigid vehicles or motorcycles unless carrying public passengers or requiring a dangerous goods driver licence – refer to definition in **Table 3**)	**Commercial standards** (Drivers of heavy vehicles, public passenger vehicles or requiring a dangerous goods driver licence – refer to definition in **Table 3**)
Hypertension	A person is **not** fit to hold an **unconditional licence**: • if the person has blood pressure consistently > 200 systolic or > 110 diastolic (treated or untreated). A **conditional licence** may be considered by the driver licensing authority subject to **periodic review**, taking into account the nature of the driving task and information provided by the **treating doctor** as to whether the following criteria are met: • the blood pressure is well controlled; **and** • there are no side effects from the medication that will impair safe driving; **and** • there is no evidence of damage to target organs relevant to driving.	A person is **not** fit to hold an **unconditional licence**: • if the person has blood pressure consistently > 170 systolic or > 100 diastolic (treated or untreated). A **conditional licence** may be considered by the driver licensing authority subject to **annual review**, taking into account the nature of the driving task and information provided by the **treating specialist*** as to whether the following criteria are met: • the person is treated with antihypertensive therapy and effective control of hypertension is achieved over a 4-week follow-up period; **and** • there are no side effects from the medication that will impair safe driving; **and** • there is no evidence of damage to target organs relevant to driving. ** Ongoing fitness to drive for commercial vehicle drivers may be assessed by the treating GP provided this is mutually agreed by the specialist, GP and driver licensing authority. The initial granting of a conditional licence must, however, be based on information provided by the specialist.*
Stroke	Refer to section 6. Neurological conditions.	Refer to section 6. Neurological conditions.

3. Diabetes mellitus

Refer also to section 2. Cardiovascular conditions*, section* 8. Sleep disorders *and section* 10. Vision and eye disorders*.*

3.1. Relevance to the driving task

3.1.1. Effects of diabetes on driving[1–3]

Diabetes may affect a person's ability to drive, either through a 'severe hypoglycaemic event' or from end-organ effects on relevant functions, including effects on vision, the heart and the peripheral nerves and vasculature of the extremities, particularly the feet. In people with type 2 diabetes, sleep apnoea is also more common (refer to section 8. Sleep disorders). The main hazard in people with insulin-treated diabetes is the unexpected occurrence of hypoglycaemia.

3.1.2. Evidence of crash risk[1,4–7]

There is a small but appreciable increase in motor vehicle crash risk for drivers with diabetes. The potential effects of hypoglycaemia are of most concern to road safety. However, findings point to a higher risk among those with a history of severe hypoglycaemia. There is also evidence that 'tighter control', as measured by the HbA1c, may be associated with increased crash risk.

3.2. General assessment and management guidelines

General management of diabetes in relation to fitness to drive is summarised in Figure 10. Note, for the purpose of the diabetes standard, appropriate specialist means an endocrinologist or consultant physician specialising in diabetes. For general guidance on diabetes management refer to relevant best practice guidelines (e.g. Royal Australasian College of General Practitioners' *Management of type 2 diabetes: a handbook for general practice*[8] or the National Health and Medical Research Council's *National evidence based clinical care guidelines for type 1 diabetes for children, adolescents and adults*[9]).

3.2.1. Hypoglycaemia

Definition: severe hypoglycaemic event[6,8,9]

For the purposes of this document, a 'severe hypoglycaemic event' is defined as an event of hypoglycaemia of sufficient severity such that the person is unable to treat the hypoglycaemia themselves and so requires someone else to administer treatment. It includes hypoglycaemia causing loss of consciousness or seizure. It can occur during driving or at any other time of the day or night. A severe hypoglycaemic event is particularly relevant to driving because it affects

brain function and may cause impairment of perception, motor skills or consciousness. It may also cause abnormal behaviour. A severe hypoglycaemic event is to be distinguished from mild hypoglycaemic events, the latter with symptoms such as sweating, tremulousness, hunger and tingling around the mouth, which are common occurrences in the life of a person with diabetes treated with insulin and some hypoglycaemic agents.

Potential causes[7]

Hypoglycaemia may be caused by many factors including non-adherence or alteration to medication, unexpected exertion, alcohol intake or irregular meals. Meal regularity and variability in medication administration may be important considerations for long-distance commercial driving or for drivers operating on shifts. Impairment of consciousness and judgement can develop rapidly and result in loss of control of a vehicle. Excessively tight control may contribute to hypoglycaemia.

Advice to drivers

The driver should be advised not to drive if a severe hypoglycaemic event is experienced while driving or at any other time, until they have been cleared to drive by a medical practitioner. The driver should also be advised to take appropriate precautionary steps to help avoid a severe hypoglycaemic event – for example, by:

- complying with general medical review requirements as requested by their general practitioner or specialist
- not driving if either their blood glucose is at or less than 5 mmol/L or if, while wearing a continuous or flash glucose monitor, the predicted glucose level is showing downward trends into the hypoglycaemia range (measured when the vehicle is parked)
- wearing a continuous or flash glucose monitor, preferably with an active hypoglycaemia alert or alarm
- not driving for more than two hours without considering having a snack
- not delaying or missing a main meal
- self-monitoring blood glucose levels before driving and every two hours during a journey, as reasonably practical
- carrying adequate glucose in the vehicle for self-treatment
- treating mild hypoglycaemia if symptoms occur while driving including:
  - safely steering the vehicle to the side of the road
  - turning off the engine and removing the keys from the ignition
  - self-treating the low blood glucose
  - checking the blood glucose levels 15 minutes or more after the hypoglycaemia has been treated and ensuring it is above 5 mmol/L
  - not recommencing driving until feeling well and until at least 30 minutes after the blood glucose is above 5 mmol/L.

Non-driving period after a 'severe hypoglycaemic event'

If a severe hypoglycaemic event occurs (as defined in section 3.2.1. Hypoglycaemia), the person should not drive for a significant period of time and will need to be urgently assessed. The minimum period of time before returning to drive is generally six weeks because it often takes many weeks for patterns of glucose control and behaviour to be re-established and for any temporary 'impaired hypoglycaemia awareness' to resolve (see below). The non-driving period will depend on factors such as identifying the reason for the episode, the specialist's opinion and the type of motor vehicle licence. The specialist's recommendation for returning to driving should be based on the patient's behaviour and objective measures of glycaemic control (documented blood glucose) over a reasonable interval.

Impaired hypoglycaemic awareness [10–14]

Impaired hypoglycaemic awareness exists when a person does not regularly sense the usual early warning symptoms of mild hypoglycaemia such as sweating, tremulousness, hunger, tingling around the mouth, palpitations and headache. It markedly increases the risk of a severe hypoglycaemic event occurring and is therefore a risk for road safety. Rates of severe hypoglycaemia may be up to seven times higher compared with those who retain hypoglycaemia awareness. Impaired hypoglycaemia awareness occurs in 20–25 per cent of people with type 1 diabetes and about 10 per cent of those with type 2 diabetes. Prevalence is higher in older people and in those with a longer duration of diabetes.

Impaired hypoglycaemic awareness may be screened for using the Clarke questionnaire (refer to Figure 9. Clarke hypoglycaemia awareness survey), which may be particularly useful for people with insulin-treated diabetes of longer duration (more than 10 years), or following a severe hypoglycaemic event or after a crash. The use of devices such as continuous or flash glucose monitors do not replace the need for a person to be able to sense the warning signs of hypoglycaemia or to compensate for impaired hypoglycaemia awareness.

When impaired hypoglycaemia awareness develops in a person who has experienced a severe hypoglycaemic event, it may improve in the subsequent weeks and months if further hypoglycaemia can be avoided.

A person with persistent impaired hypoglycaemia awareness should be under the regular care of a medical practitioner with expert knowledge in managing diabetes (e.g. an endocrinologist or diabetes specialist), who should be involved in assessing their fitness to drive. As reflected in the standards table on page 101, any driver who has a persistent impaired hypoglycaemia awareness is generally not fit to drive unless their ability to experience early warning symptoms returns or they have an effective management strategy for lack of early warning symptoms. For private drivers, a conditional licence may be considered by the driver licensing authority, taking into account the opinion of an appropriate specialist, the nature and extent of the driving involved and the driver's self-care behaviours.

In managing impaired hypoglycaemia awareness, the medical practitioner should focus on aspects of the person's self-care to minimise a severe hypoglycaemic event occurring while driving, including steps described above (*Advice to drivers*). In addition, self-care behaviours that help to minimise severe hypoglycaemic events in general should be a major ongoing focus of regular diabetes care. This requires attention by both the medical practitioner and the person with diabetes to diet and exercise approaches, insulin regimens and blood glucose testing protocols.

3.2.2. Acute hyperglycaemia

While acute hyperglycaemia may affect some aspects of brain function, there is not enough evidence to determine the regular effects on driving performance and related crash risk. Each person with diabetes should be counselled about managing their diabetes during days when they are unwell and should be advised not to drive if they are acutely unwell with metabolically unstable diabetes.

3.2.3. Comorbidities and end-organ complications[1–3,8,9]

Assessment and management of comorbidities is an important aspect of managing people with diabetes with respect to their fitness to drive. This should be part of routine review as per recommended practice and may include, but is not limited to, the following.

Vision

Refer to section 10. Vision and eye disorders. Visual acuity should be tested annually. Retinal screening should be undertaken every second year if there is no retinopathy, or more frequently if at high risk. Visual field testing is not required unless clinically indicated.[7,8]

Neuropathy and foot care

While it can be difficult to be prescriptive about neuropathy in the context of driving, it is important that the severity of the condition is assessed. Adequate sensation and movement for the operation of foot controls is required (refer to section 6. Neurological conditions and section 5. Musculoskeletal conditions).

Sleep apnoea

Sleep apnoea is a common comorbidity affecting many people with type 2 diabetes and has substantial implications for road safety. The treating health professional should be alert to potential signs (e.g. BMI greater than 35) and symptoms, and apply the Epworth Sleepiness Scale as appropriate (refer to section 8. Sleep disorders).

Cardiovascular

There are no diabetes-specific medical standards for cardiovascular risk factors and driver licensing. Consistent with good medical practice, people with diabetes should have their cardiovascular risk factors periodically assessed and treated as required (refer to section 2. Cardiovascular conditions).

3.2.4. Gestational diabetes mellitus

The standards in this chapter apply to diabetes mellitus as a chronic condition. The self-limiting condition known as gestational diabetes mellitus does not affect licensing. However, consideration should be given to short-term fitness to drive in women with gestational diabetes mellitus treated with insulin, although severe hypoglycaemia in this condition is rare. Affected women should be counselled to recognise symptoms and to restrict driving when symptoms occur.

Figure 9. Clarke hypoglycaemia awareness survey[10]

Patient Survey

The survey is useful to administer to assess hypoglycaemia awareness including for people who have been on insulin for many years after a severe hypoglycaemic event following a crash.

Scoring

- Four or more 'R' responses implies reduced awareness.
- For questions 5 and 6, one 'R' response is given if the answer to question 5 is less than the answer to question 6.
- 'A' responses imply awareness.
- 'U' response (12 or more severe hypoglycaemic episodes in the last 12 months) indicates unawareness.

1. Check the category that best describes you: (check one only)

   - ☐ I always have symptoms when my blood sugar is low (A).
   - ☐ I sometimes have symptoms when my blood sugar is low (R).
   - ☐ I no longer have symptoms when my blood sugar is low (R).

2. Have you lost some of the symptoms that used to occur when your blood sugar was low?

   - ☐ Yes (R)
   - ☐ No (A)

3. In the past six months, how often have you had moderate hypoglycaemia episodes? (episodes where you might feel confused, disoriented or lethargic and were unable to treat yourself)

   - ☐ Never (A)
   - ☐ Once a month (R)
   - ☐ Once or twice (R)
   - ☐ More than once a month (R)
   - ☐ Every other month (R)

4. In the past year, how often have you had severe hypoglycaemic episodes? (episodes where you were unconscious or had a seizure and needed glucagon or intravenous glucose)

   - ☐ Never (A)
   - ☐ 1 time (R)
   - ☐ 2 times (R)
   - ☐ 3 times (R)
   - ☐ 5 times (R)
   - ☐ 6 times (R)
   - ☐ 7 times (R)
   - ☐ 8 times (R)
   - ☐ 9 times (R)
   - ☐ 10 times (R)
   - ☐ 11 times (R)
   - ☐ 12 or more times (U)

3.3. Medical standards for licensing

Medical requirements for unconditional and conditional licences are outlined in the table on page 101. Health professionals should familiarise themselves with the information in this chapter and the tabulated standards before assessing a person's fitness to drive.

3.3.1. Diabetes treated by glucose-lowering agents other than insulin for private drivers

Private vehicle drivers treated by glucose-lowering agents other than insulin may generally drive without licence restriction (i.e. on an unconditional licence) but should be required by the driver licensing authority to have five-yearly reviews.

3.3.2. Recommendation and review of conditional licences for commercial vehicle drivers

It is a general requirement that conditional licences for commercial vehicle drivers are issued by the driver licensing authority based on advice from an appropriate medical specialist (endocrinologist or consultant physician specialising in diabetes) and that these drivers are reviewed periodically by the specialist to determine their ongoing fitness to drive (refer to Part A section 4.4. Conditional licences). For commercial drivers receiving insulin treatment, at least three months of blood glucose monitoring records should be reviewed in assessing fitness to drive.

Commercial vehicle drivers treated by glucose-lowering agents other than insulin must have at least an annual review by an appropriate specialist to monitor the progression of their condition. However, in the case of type 2 diabetes managed by metformin alone, ongoing fitness to drive may be assessed by the treating general practitioner by mutual agreement with the specialist. The initial recommendation of a conditional licence must be based on the opinion of an endocrinologist or consultant physician specialising in diabetes.

In areas where access to specialists may be difficult, the driver licensing authority may agree to a process in which:

- initial assessment and advice for the conditional licence is provided by a specialist (endocrinologist or consultant physician specialising in diabetes)
- ongoing periodic review for the conditional licence is provided by the treating general practitioner, with the cooperation of the specialist.

Where appropriate and available, the use of telemedicine technologies such as videoconferencing is encouraged as a means of facilitating access to specialist opinion (refer to Part A section 3.3.5. Role of medical specialists).

Medical standards for licensing – diabetes mellitus Health professionals should familiarise themselves with the information in this chapter and the tabulated standards before assessing a person's fitness to drive.		
Condition	**Private standards** (Drivers of cars, light rigid vehicles or motorcycles unless carrying public passengers or requiring a dangerous goods driver licence – refer to definition in **Table 3**)	**Commercial standards** (Drivers of heavy vehicles, public passenger vehicles or requiring a dangerous goods driver licence – refer to definition in **Table 3**)
Diabetes controlled by diet and exercise alone	A person with diabetes treated by diet and exercise alone may drive without licence restriction. They should be reviewed by their treating doctor periodically regarding the progression of their diabetes.	A person with diabetes treated by diet and exercise alone may drive without licence restriction. They should he reviewed by their treating doctor periodically regarding the progression of their diabetes.
Diabetes treated by glucose-lowering agents other than insulin For definition and management of a 'severe hypoglycaemic event' refer to section 3.2.1	A person is **not** fit to hold an **unconditional licence**: • if the person has end-organ complications that may affect driving, as per this publication; or • the person has had a recent 'severe hypoglycaemic event'. A **conditional licence** may be considered by the driver licensing authority subject to **periodic review**, taking into consideration the nature of the driving task and information provided by the **treating doctor** on whether the following criteria are met: • any end-organ effects are satisfactorily treated, with reference to the standards in this publication; **and** • the person is following a treatment regimen that minimises the risk of hypoglycaemia; **and** • the person experiences early warning symptoms (awareness) of hypoglycaemia or has a documented management plan for lack of early warning symptoms; **and** • any recent 'severe hypoglycaemic event' has been satisfactorily treated, with reference to the standards in this publication (refer to section 3.2.1). For private drivers who do not meet the above criteria, a **conditional licence** may be considered by the driver licensing authority, taking into account the opinion of an **endocrinologist or consultant physician specialising in diabetes** and subject to regular specialist review.	A person is **not** fit to hold an **unconditional licence**: • if the person has non–insulin treated diabetes mellitus and is being treated with glucose-lowering agents other than insulin. A **conditional licence** may be considered by the driver licensing authority subject to at least **annual review**, taking into consideration the nature of the driving task and information provided by an **endocrinologist or consultant physician specialising in diabetes*** on whether the following criteria are met: • there is no recent history of a 'severe hypoglycaemic event' as assessed by the specialist; **and** • the person experiences early warning symptoms (awareness) of hypoglycaemia; **and** • the person is following a treatment regimen that minimises the risk of hypoglycaemia; **and** • there is an absence of end-organ effects that may affect driving as per this publication. ** For a commercial driver with type 2 diabetes who is being treated with metformin alone, the annual review for a conditional licence may be undertaken by the driver's treating doctor upon mutual agreement of the treating doctor, specialist and driver licensing authority. The initial granting of a conditional licence must, however, be based on information provided by the specialist.*

Medical standards for licensing – diabetes mellitus		
Condition	**Private standards** (Drivers of cars, light rigid vehicles or motorcycles unless carrying public passengers or requiring a dangerous goods driver licence – refer to definition in **Table 3**)	**Commercial standards** (Drivers of heavy vehicles, public passenger vehicles or requiring a dangerous goods driver licence – refer to definition in **Table 3**)
Insulin-treated diabetes (except gestational diabetes) For definition and management of a 'severe hypoglycaemic event' refer to section 3.2.1	A person is **not** fit to hold an **unconditional licence**: • if the person has insulin-treated diabetes. A **conditional licence** may be considered by the driver licensing authority subject to at least **2-yearly review**, taking into consideration the nature of the driving task and information provided by the **treating doctor** on whether the following criteria are met: • there is no recent history of a 'severe hypoglycaemic event'; **and** • the person is following a treatment regimen that minimises the risk of hypoglycaemia; **and** • the person experiences early warning symptoms (awareness) of hypoglycaemia (refer to section 3.2.1) or has a documented management plan for lack of early warning symptoms; **and** • there are no end-organ effects that may affect driving as per this publication. For private drivers who do not meet the above criteria, a **conditional licence** may be considered by the driver licensing authority, taking into account the opinion of an **endocrinologist or consultant physician specialising in diabetes** and subject to regular specialist review.	A person is **not** fit to hold an **unconditional licence**: • if the person has insulin-treated diabetes. A **conditional licence** may be considered by the driver licensing authority subject to at least **annual review**, taking into consideration the nature of the driving task and information provided by an **endocrinologist or consultant physician specialising in diabetes** on whether the following criteria are met: • there is no recent history (generally at least **6 weeks**) of a 'severe hypoglycaemic event' as assessed by the specialist; **and** • the person is following a treatment regimen that minimises the risk of hypoglycaemia; **and** • the person experiences early warning symptoms (awareness) of hypoglycaemia (refer to section 3.2.1); **and** • there are no end-organ effects that may affect driving as per this publication.

IMPORTANT: The medical standards and management guidelines contained in this chapter should be read in conjunction with the general information contained in Part A of this publication. Practitioners should give consideration to the following:

Licensing responsibility

The responsibility for issuing, renewing, suspending or cancelling a person's driver licence (including a conditional licence) lies ultimately with the driver licensing authority.

Licensing decisions are based on a full consideration of relevant factors relating to health and driving performance.

Conditional licences

For a conditional licence to be issued, the health professional must provide to the driver licensing authority details of the medical criteria not met, evidence of the medical criteria met, as well as the proposed conditions and monitoring requirements.

The presence of other medical conditions

While a person may meet individual disease criteria, concurrent medical conditions may combine to affect fitness to drive – for example, hearing, visual or cognitive impairment (refer to Part A section 2.2.7. Older drivers and age-related changes and section 2.2.8. Multiple medical conditions).

The nature of the driving task

The driver licensing authority will take into consideration the nature of the driving task as well as the medical condition, particularly when granting a conditional licence. For example, the licence status of a farmer requiring a commercial vehicle licence for the occasional use of a heavy vehicle may be quite different from that of an interstate multiple combination vehicle driver. The examining health professional should bear this in mind when examining a person and when providing advice to the driver licensing authority.

Reporting responsibilities

Patients should be made aware of the effects of their condition on driving and should be advised of their legal obligation to notify the driver licensing authority where driving is likely to be affected. The health professional may themselves advise the driver licensing authority as the situation requires (refer to section 3.3 and step 6).

4. Hearing loss and deafness

Refer also to Part B section 5. Musculoskeletal conditions *and section* 10. Vision and eye disorders.

This section deals with fitness to drive in relation to hearing loss and deafness. Hearing loss and deafness may be well compensated for when driving by relying on vision, attention and physical mobility to adequately scan the driving environment. Hearing loss and deafness may coexist with other impairments that could compromise these adaptive capabilities, such as visual, mobility and cognitive impairment, particularly in older people. For guidance in assessing multiple medical conditions refer to Part A 2.2.8. Multiple medical conditions.

4.1. Relevance to the driving task

4.1.1. Effect of hearing loss on driving[1–7]

The evidence base regarding hearing loss and driving safely is limited; however, the analysis of the small number of quality studies report that hearing loss has no effect on motor vehicle crash risk.[1] It may be that a loss of hearing is well compensated for since most people who are hard of hearing are aware of their disability and therefore tend to be more cautious and rely more on visual cues and other sensations such as vibrations. Functional hearing is used in the driving task to:

- sense alarms, notifications and other auditory cues in the driving environment (e.g. railway crossings, first responder sirens)
- sense auditory cues on the state and position of the vehicle from vehicle technology systems (e.g. alarms, warning systems using sound alerts)
- compensate for comorbidities or general reduction in functional abilities that may impair driving (e.g. slower reaction times, reduced neck rotation due to musculoskeletal changes)
- assist in maintaining general vigilance, alertness and arousal.

4.1.2. Considerations for commercial vehicle drivers[8,9]

While driving ability per se might not be affected by a hearing loss, responsiveness to critical events is an important safety consideration for drivers of commercial vehicles. These drivers therefore require the capacity to ensure safety and the capacity to respond to environmental situations that may involve sirens, rail crossings and emergency signals as well as conditions of the vehicle and roads. It may be challenging to rely solely on vision to compensate for a significant hearing loss in a commercial vehicle. The internal commercial vehicle cabin can place additional attentional, workload and visual demands on drivers for undertaking tasks such as checking multiple displays or monitors and using communication systems. These additional tasks reduce capacity to compensate for a lack of hearing by relying on vision to monitor the external environment.

4.2. General assessment and management guidelines

4.2.1. Commercial vehicle drivers

Only drivers of commercial vehicles are required to meet a hearing standard for the reasons outlined above. The following hearing assessment applies to all forms of hearing loss including congenital and childhood hearing loss, and hearing loss acquired in later years. The process is summarised in Figure 11. Management of hearing loss in commercial vehicle drivers.

- Compliance with the standard should be clinically assessed initially by the treating doctor, audiologist* or audiometrist**. If there is doubt about the person's hearing, audiometry should be performed by an ear, nose and throat (ENT) specialist, audiologist* or audiometrist**. The person need not undergo audiometry if their hearing is satisfactory.
- If on audiometry the person has unaided hearing loss greater than or equal to 40 dB in the better ear (averaged over the frequencies 0.5, 1, 2 and 3 KHz), they do not meet the criteria for an unconditional licence.
- If the standard is able to be met with a hearing aid, the driver licensing authority may consider a conditional licence, subject to periodic assessment of hearing, the use of the hearing aid whilst driving and of the hearing aids set at a frequency advised by the ENT specialist, audiologist* or audiometrist**. Stable conditions may not require periodic review.
- If the standard is not able to be met with a hearing aid, this does not disqualify the person from driving. They should be offered individualised assessment to determine their eligibility for a conditional licence. This may comprise assessment by an ENT specialist or audiologist* including consideration of:
  - the person's medical history – for example, childhood deafness may have led to good adaptation
  - the person's driving record before and since the hearing loss
  - the nature of the driving task – for example, the type of vehicle (e.g. truck, bus), roads and distances to be travelled
  - the ergonomics of the driving cab – for example, assistive devices such as mirrors and a GPS
  - concomitant medical conditions such as vision impairment or cognitive impairment
  - practical driver assessment if required (refer to Part A section 2.3.1. Practical driver assessments). The report may advise on assistive technologies as a licence condition.

The driver licensing authority may consider a conditional licence based on the information received. Periodic review may include medical review and/or a practical driver assessment at the discretion of the driver licensing authority. The health professional should advise on the frequency of review as determined by the natural history of the condition. Stable conditions may not require periodic review.

In some cases, noise amplification from wearing hearing aids may lead to driver distraction and may warrant individualised assessment as above to determine fitness to drive without the hearing aid.

4.2.2. Private vehicle drivers

While hearing loss will not preclude driving a private car, people with hearing loss should be advised regarding their loss and their limited ability to hear warning signals. Assistive technologies such as hearing aids, sensors and/or physical equipment such as additional mirrors might also be used upon consideration of the needs of the individual driver. Occupational therapist driving assessors can assist with identifying vehicle aides.

* For the purposes of this document an audiologist is a person accredited as such by Audiology Australia (refer to www.audiology.asn.au) or the Australian College of Audiology (refer to www.acaud.org).

** For the purposes of this document, an audiometrist is a person accredited as such by the Australian College of Audiology (refer to www.acaud.org) or the Hearing Aid Audiology Society of Australia (refer to www.haasa.org.au).

4.3. Medical standards for licensing

Requirements for unconditional and conditional licences are outlined in the following table.

Medical standards for licensing – hearing **Health professionals should familiarise themselves with the information in this chapter and the tabulated standards before assessing a person's fitness to drive.**		
Condition	**Private standards** (Drivers of cars, light rigid vehicles or motorcycles unless carrying public passengers or requiring a dangerous goods driver licence – refer to definition in **Table 3**)	**Commercial standards** (Drivers of heavy vehicles, public passenger vehicles or requiring a dangerous goods driver licence – refer to definition in **Table 3**)
Hearing loss	There is no hearing standard for private vehicle drivers. Refer to 4.2. General assessment and management guidelines.	Compliance with the standard should be clinically assessed initially. If the initial clinical assessment indicates possible hearing loss, the person should be referred for audiometry. A person is **not** fit to hold an **unconditional licence**: • if the person has unaided hearing loss ≥ 40 dB in the better ear (averaged over the frequencies 0.5, 1, 2 and 3 KHz). A **conditional licence** may be considered by the driver licensing authority subject to **periodic review***, taking into account the nature of the driving task and information provided by an **ENT specialist or audiologist****, as to whether: • the standard is able to be met with a hearing aid***. If the standard is **not** able to be met with a hearing aid, further individualised assessment should be offered. A **conditional licence** may be considered by the driver licensing authority subject to **periodic review***, taking into account: • the nature of the driving task; **and** • information provided by an ENT specialist or audiologist**; **and** • the results of a practical driver assessment if required. ** Stable conditions may not require periodic review.* *** Refer to section 4.2. General assessment and management guidelines.* **** In some cases, noise amplification as a result of wearing hearing aids may lead to driver distraction and may warrant individualised assessment to determine fitness to drive without the hearing aid (refer to 4.2. General assessment and management guidelines).*

IMPORTANT: The medical standards and management guidelines contained in this chapter should be read in conjunction with the general information contained in Part A of this publication. Practitioners should give consideration to the following:

Licensing responsibility

The responsibility for issuing, renewing, suspending or cancelling a person's driver licence (including a conditional licence) lies ultimately with the driver licensing authority.

Licensing decisions are based on a full consideration of relevant factors relating to health and driving performance.

Conditional licences

For a conditional licence to be issued, the health professional must provide to the driver licensing authority details of the medical criteria not met, evidence of the medical criteria met, as well as the proposed conditions and monitoring requirements.

The presence of other medical conditions

While a person may meet individual disease criteria, concurrent medical conditions may combine to affect fitness to drive – for example, hearing, visual or cognitive impairment (refer to Part A section 2.2.7. Older drivers and age-related changes and section 2.2.8. Multiple medical conditions).

The nature of the driving task

The driver licensing authority will take into consideration the nature of the driving task as well as the medical condition, particularly when granting a conditional licence. For example, the licence status of a farmer requiring a commercial vehicle licence for the occasional use of a heavy vehicle may be quite different from that of an interstate multiple combination vehicle driver. The examining health professional should bear this in mind when examining a person and when providing advice to the driver licensing authority.

Reporting responsibilities

Patients should be made aware of the effects of their condition on driving and should be advised of their legal obligation to notify the driver licensing authority where driving is likely to be affected. The health professional may themselves advise the driver licensing authority as the situation requires (refer to section 3.3 and step 6).

10. Vision and eye disorders

This chapter focuses on the assessment methods, medical criteria and management approach for the two main aspects of vision – visual acuity and visual fields. It should be read in conjunction with other chapters where those conditions may affect vision (e.g, neurological conditions (section 6.3. Other neurological and neurodevelopmental conditions), diabetes (section 3. Diabetes mellitus) and with Part A section 2.2.7. Older drivers and age-related changes.

10.1. Relevance to the driving task

10.1.1. Effects of vision and eye disorders on driving[1–6]

Good vision, including visual acuity and visual fields, is essential to operating a motor vehicle. Any marked loss of visual acuity or visual fields will diminish the person's ability to drive safely, including their ability to detect another vehicle, pedestrians or warning signs. It may also increase the time for a person to perceive and react to a potentially hazardous situation.

Peripheral or side vision assists the driver to be aware of the total driving environment and is particularly important in certain common driving tasks, such as merging into a traffic stream or changing lanes, and in detecting pedestrians and vehicles to the side of the line of vision.

Vision defects can develop slowly, and drivers may be unaware of their reduced abilities, particularly in relation to peripheral vision.

10.1.2. Evidence of crash risk[1–3,7,8,15,16]

The evidence is incomplete regarding visual fields and visual acuity and crash risk. This is likely due to the many methodological reasons outlined in Part A of this publication (refer to Part A section 1.5. Development and evidence base).

Identifying the degree to which reduced visual acuity increases motor vehicle crash risk is challenging. The evidence for visual acuity impairment is generally limited to drivers whose visual acuity already meets licensing standards, limiting the ability to determine whether alternative cut-points are more appropriate. While it is generally agreed that adequate visual fields are important for safe driving, the actual cut-off value that should be set remains unclear.

Most research suggests there is no association between crash risk and colour vision. While there is evidence that people with red-colour-deficient vision have difficulty in detecting red lights and stopping in laboratory and on-road testing, significant improvements in road engineering mean that people with red-colour deficiency may largely compensate for their deficiency while driving.

10.2. General assessment and management guidelines

Decline in vision is associated with normal ageing and is therefore an important consideration for fitness to drive in the general care of older people, along with consideration of cognition and sensory-motor function (refer to Part A section 2.2.7. Older drivers and age-related changes).

Progressive eye conditions such as cataracts, glaucoma and macular degeneration are also more common in older people. Once diagnosed, these conditions require regular monitoring in relation to driving, including through conditional licences as appropriate (refer to section 10.2.4. Progressive eye conditions). Regular monitoring is also required for conditions such as diabetes to screen for and manage end-organ effects (retinopathy) (refer to section 3.2.3. Comorbidities and end-organ complications).

For drivers with neurological conditions such as stroke, vision is one of a number of functional outcomes that will be addressed as part of an overall assessment of fitness to drive, and findings will need to be integrated as part of this overall assessment (refer to section 6.3. Other neurological and neurodevelopmental conditions).

10.2.1. Visual acuity[3–6]

For the purposes of this publication, visual acuity is defined as a person's clarity of vision with or without glasses or contact lenses. Where a person does not meet the visual acuity standard at initial assessment, they may be referred for further assessment by an optometrist or ophthalmologist.

Assessment method

Visual acuity should be measured for each eye separately and without optical correction. If optical correction is needed, vision should be retested with appropriate corrective lenses. For use of orthokeratology lenses to correct visual acuity, refer to section 10.2.7. Orthokeratology therapy.

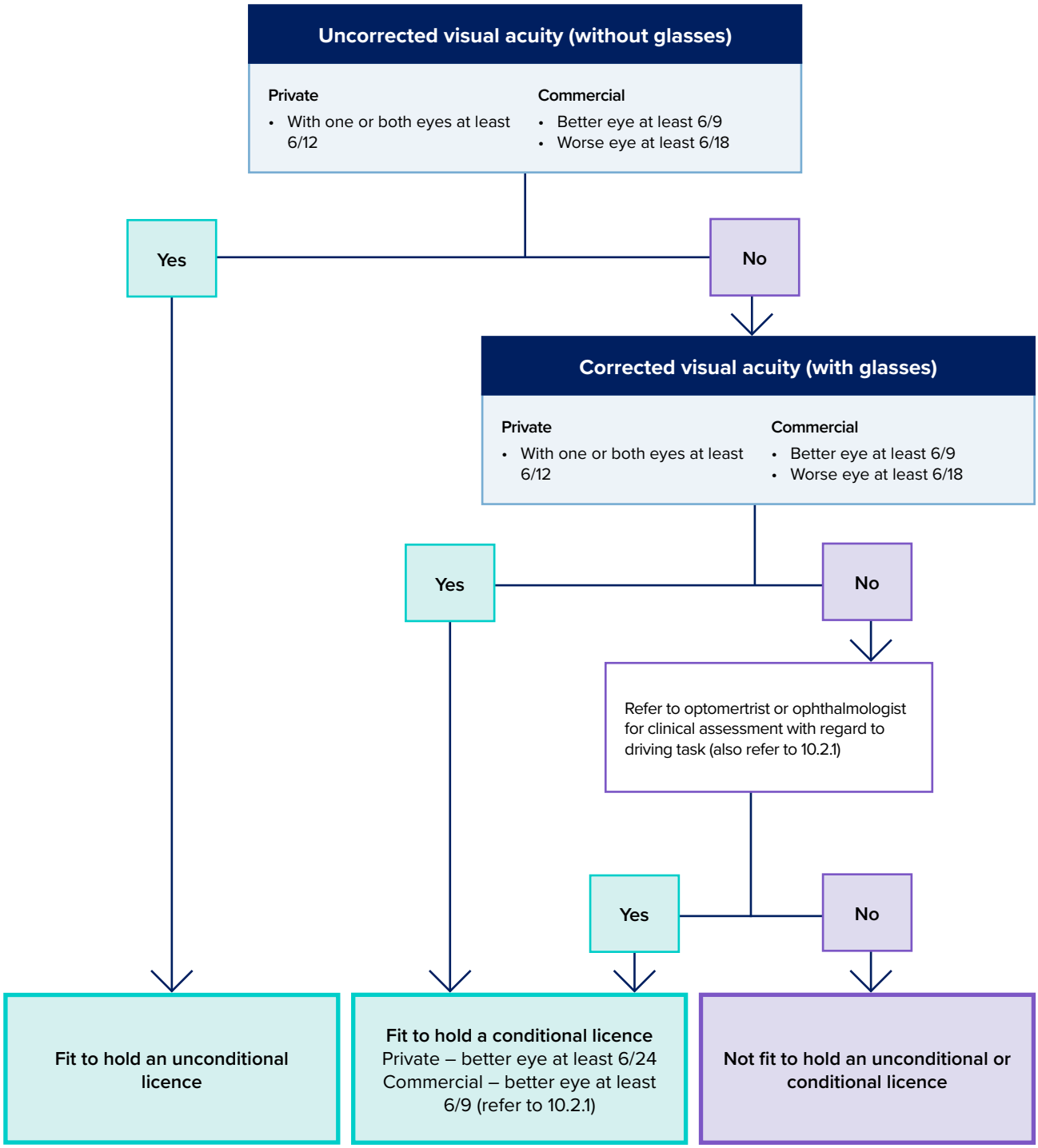
Acuity should be tested using a standard visual acuity chart (Snellen or LogMAR chart or equivalent) with five letters on the 6/12 line. Standard charts should be placed six metres from the person tested; otherwise, a reverse chart can be used and viewed through a mirror from a distance of three metres. Other calibrated charts can be used at a minimum distance of three metres. More than two errors in reading the letters of any line is regarded as a failure to read that line. Refer to Figure 17 for a management flow chart.

In the case of a private vehicle driver, if the person's visual acuity is just below that required by the standard but the person is otherwise alert, has normal reaction times and good physical coordination, an optometrist or ophthalmologist can recommend the granting of a conditional licence. The use of contrast sensitivity or other specialised tests may help in the assessment. However, a driver licence will not be issued when visual acuity in the better eye is worse than 6/24 for private vehicle drivers.

There is also some flexibility for commercial vehicle drivers depending on the driving task, providing the visual acuity in the driver's better eye (with or without corrective lenses) is 6/9 or better.

Restrictions on driving (conditional licence) may be advised; for example, where glare is a marked problem, no-night driving may be recommended. Refer to Part A section 4.4. Conditional licences.

Figure 17. Visual acuity requirements for private and commercial vehicle drivers



10.2.2. Visual fields[3–6,8–13]

For the purposes of this publication, visual fields are defined as a measure of the extent of peripheral (side) vision. Normal visual field is: 60 degrees nasally, 100 degrees temporally, 75 degrees inferiorly and 60 degrees superiorly. The binocular field extends the horizontal extent from 160 to 200 degrees, with the central 120 degrees overlapping and providing the potential for stereopsis.

Visual fields may be reduced due to a range of neurological conditions (e.g. stroke, multiple sclerosis) as well by ocular diseases (e.g. glaucoma), or injuries, resulting in hemionopia, quadrantanopia or monocularity.

Peripheral vision assists the driver to be aware of the total driving environment. Once alerted, the central fovea area is moved to identify the importance of the information. Therefore, peripheral vision loss that is incomplete will still allow awareness; this includes small areas of loss and patchy loss. Additionally, affected drivers can adapt to the defect by scanning regularly and effectively and can have good awareness. Patients with visual field defects who have full intellectual/cognitive capacity are more able to adapt, but those with such impairments will have decreased awareness and are therefore not safe to drive.

A longstanding field defect, such as from childhood, may lead to visual adaptation. Such defects need to be assessed by an optometrist or ophthalmologist for a conditional licence to be considered. They should be managed as an exceptional case to the standard, with consideration for duration and evidence of visual adaptation, whether the location of the defect is an area that may already be blocked by the car door on the passenger side (i.e. the inferior field on the left side without central field loss), driver safety record and the nature of the driving task.

Assessment method

If there is no clinical indication of a visual field impairment or a progressive eye condition, then it is satisfactory to screen for defect by confrontation. Confrontation is an inexact test. Any person who has, or is suspected of having, a visual field defect should have a formal perimetry-based assessment.

Monocular automated static perimetry is the minimum baseline standard for visual field assessments. If monocular automated static perimetry shows no visual field defect, this information is sufficient to confirm that the standard is met.

Subjects with any significant field defect or a progressive eye condition require a binocular Esterman visual field for assessment. This is classically done on a Humphrey visual field analyser, but any machine that can be shown to be equivalent is accepted (e.g. Medmont binocular VF printed off in level map mode). The treating optometrist or ophthalmologist can determine whether it is appropriate for the person to wear their normal corrective lenses while undergoing testing. Fixation monitoring must be performed and recorded on the test. Alternative devices must have the ability to monitor fixation and to stimulate the same spots as the standard binocular Esterman. For an Esterman binocular chart to be considered reliable for licensing, the false-positive score must be no more than 20 per cent.

Horizontal extent of the visual field

In the case of a private vehicle driver, if the horizontal extension of a person's visual fields are less than 110 degrees but greater or equal to 90 degrees, an optometrist or ophthalmologist may support the granting of a conditional licence by the driver licensing authority. The extent is measured on the Esterman from the last seen point to the next seen point. There is no flexibility in this regard for commercial vehicle drivers.

A single cluster of up to three adjoining missed points, unattached to any other area of defect, lying on or across the horizontal meridian will be disregarded when assessing the horizontal extension of the visual field. A vertical defect of only a single point width but of any length, unattached to any other area of defect, that touches or cuts through the horizontal meridian may be disregarded. There should be no significant defect in the binocular field that encroaches within 20 degrees of fixation above or below the horizontal meridian. This means that homonymous or bitemporal defects that come close to fixation, whether hemianopic or quadrantanopic, are not normally accepted as safe for driving.

Central field loss

Scattered single missed points or a single cluster of up to three adjoining points is acceptable central field loss for a person to hold an unconditional licence. **A significant or unacceptable central field loss is defined as any of the following:**

- a cluster of four or more adjoining points that is either completely or partly within the central 20-degree area
- loss consisting of both a single cluster of three adjoining missed points up to and including 20 degrees from fixation, and any additional separate missed point(s) within the central 20-degree area
- any central loss that is an extension of a hemianopia or quadrantanopia of size greater than three missed points.

Methods of measuring visual fields are limited in their ability to resemble the demands of the real-world driving environment where drivers are free to move their eyes as required and must sustain their visual function in variable conditions. Additional factors to be considered by the driver licensing authority in assessing patients with defects in visual fields therefore include, but are not limited to, the following:

- kinetic fields conducted on a Goldman
- binocular Esterman visual fields conducted without fixation monitoring, often referred to as a roving Esterman (two consecutive tests must be performed with no more than one false-positive allowed) – the test should be in the numeric field format when it is printed out or sent for an opinion
- contrast sensitivity and glare susceptibility
- medical history; duration and prognosis; if the condition is progressive; rate of progression/deterioration; effectiveness of treatment/management
- driving record before and since the occurrence of the defect
- the nature of the driving task – for example, type of vehicle (truck, bus, etc.), roads and distances to be travelled concomitant medical conditions such as cognitive impairment or impaired rotation of the neck.

There is no flexibility in this regard for commercial vehicle drivers.

Monocular vision (one-eyed driver)

Monocular drivers have a reduction of visual fields due to the nose obstructing the medial visual field. They also have no stereoscopic vision and may have other deficits in visual functions.

For private vehicle drivers, a conditional licence may be considered by the driver licensing authority if the horizontal visual field is 110 degrees and the visual acuity is satisfactory in the better eye. People with monocular vision are generally not fit to drive a commercial vehicle. A conditional licence may be considered by the driver licensing authority if the horizontal visual field is 140 degrees, the visual acuity in the better eye is satisfactory, there is no other visual field loss that is likely to impede driving and an ophthalmologist/optometrist assesses that the person may be safe to drive after consideration of the above factors. The better eye must be reviewed at least every two years.

If monocular automated static perimetry is undertaken on patients without symptoms, family history or risk factors for visual field loss, and shows no indication of any visual field concerns, this information may be sufficient to confirm that the standard is met. If monocular testing suggests a field defect, or if the patient has a progressive eye condition, and/or the patient has any other symptoms or signs that indicate a field defect, then binocular testing should be conducted using the Esterman binocular field test or an Esterman-equivalent test. Alternative devices must have the ability to monitor fixation and to stimulate the same spots as the standard binocular Esterman.

Sudden loss of unilateral vision

A person who has lost an eye or most of the vision in an eye on a long-term basis has to adapt to their new visual circumstances and re-establish depth perception. They should therefore be advised not to drive for an appropriate period after the onset of their sudden loss of vision (usually three months). They should notify the driver licensing authority and be assessed according to the relevant visual field standard.

10.2.3. Diplopia

People suffering from all but minor forms of diplopia are generally not fit to drive. Any person who reports or is suspected of experiencing diplopia within 20 degrees from central fixation should be referred for assessment by an optometrist or ophthalmologist. For diplopia managed with an occluder, a three-month non-driving period applies in order to re-establish depth perception.

10.2.4. Progressive eye conditions[3]

The patient should be advised appropriately when a progressive eye condition is diagnosed that may result in future restrictions on driving. It is important to give the patient as much lead time as possible to prepare for changes that may later be required (e.g. adaptation to alternate transport and/or engaging blindness and low vision services). Refer to Part A section 2.2.6. Congenital conditions, disability and driving and 2.2.7. Older drivers and age-related changes.

People with progressive eye conditions such as cataract, glaucoma, optic neuropathy, diabetic retinopathy, macular degeneration or retinitis pigmentosa should be monitored regularly and should be advised in advance about the potential future impact on their driving ability.

10.2.5. Congenital and acquired nystagmus

Nystagmus may reduce visual acuity. Drivers with nystagmus must meet the visual acuity standard. Any underlying condition must be fully assessed to ensure there is no other issue that relates to fitness to drive. Those who have congenital nystagmus may have developed coping strategies that are compatible with safe driving and should be individually assessed by an appropriate specialist.

10.2.6. Colour vision

There is not a colour vision standard for drivers, either private or commercial. Doctors, optometrists and ophthalmologist should, however, advise drivers who have a significant colour vision deficiency about how this may affect their responsiveness to signal lights and the need to adapt their driving accordingly. Note, this standard applies only to driving within normal road rules and conditions. A standard requiring colour vision may be justified based on risk assessment for particular driving tasks.

10.2.7. Orthokeratology therapy[14]

Orthokeratology involves the therapeutic use of rigid gas-permeable contact lenses worn overnight to reshape the cornea of the eye. This provides effective correction of visual refractive error (once the lenses are removed) that can last at least a full day. The therapeutic effect is temporary and so the lenses must be worn regularly to maintain the best visual outcomes.

A conditional licence can be considered for private and commercial vehicle drivers provided the visual acuity standard is met with orthokeratology therapy and the lenses are worn as recommended by an optometrist or ophthalmologist. The driver may drive without their normal correcting lenses (e.g. glasses or contact lens) provided that the visual acuity standard is maintained with the support of orthokeratology therapy. If the driver cannot meet or maintain the standard using orthokeratology therapy, they must drive with correcting lenses that enable them to meet the standard.

10.2.8. Telescopic lenses (bioptic telescopes)

> The driver licensing authority may refuse a licence if the visual acuity standards are not met without the use of a bioptic telescope (refer to section 10.2.1. Visual acuity). People seeking to use a bioptic device for driving should first contact their driver licensing authority and check whether these devices are an accepted means to meet the standards. Refer to Appendix 9 for driver licensing authority contact details.

Bioptic telescopes are devices used to compensate for reduced visual acuity. They are miniature telescopes typically mounted on the upper part of a person's glasses. Bioptics are used momentarily and intermittently when driving, the majority of which occurs at the corrected visual acuity provided by the person's glasses. The person drops their chin slightly to view through the telescope for magnification, then lifts their chin to view through their standard corrective lens.

At present, there is insufficient information from human factors and safety research of drivers using these devices to set standards for bioptics. As such, and due to the increased risk associated with commercial vehicle driving (refer to Part A section 4.1. Medical standards for private and commercial vehicle drivers), these devices should not be used to meet the visual acuity standards for commercial vehicles. For private vehicle drivers, the driver licensing authority may consider information from an

assessment performed by an ophthalmologist or optometrist when making its licensing decision.

10.2.9. Practical driver assessments

A practical driver assessment is not considered to be a safe or reliable method of assessing the effects of disorders of vision on driving, especially the visual fields, because the driver's response to emergency situations or various environmental conditions cannot be determined. Information about adaptation to visual field defects can be gained from visual field tests such as the Esterman.

A practical driver assessment may be helpful in assessing the ability to process visual information (refer to Part A section 2.3.1. Practical driver assessments).

10.2.10. Exceptional cases

In unusual circumstances, cases may be referred by the driver licensing authority for further medical specialist opinion (refer to Part A section 3.3.7. Role of independent experts/panels).

10.3. Medical standards for licensing

Requirements for unconditional and conditional licences are outlined in the following tables.

Medical standards for licensing – vision and eye disorders

Health professionals should familiarise themselves with the information in this chapter and the tabulated standards before assessing a person's fitness to drive.

Condition	Private standards (Drivers of cars, light rigid vehicles or motorcycles unless carrying public passengers or requiring a dangerous goods driver licence – refer to definition in **Table 3**)	Commercial standards (Drivers of heavy vehicles, public passenger vehicles or requiring a dangerous goods driver licence – refer to definition in **Table 3**)
Visual acuity Refer to section 10.2.1. Visual acuity and Figure 17.	A person is **not** fit to hold an **unconditional licence**: • if the person's uncorrected visual acuity in the better eye or with both eyes together is worse than 6/12. A **conditional licence** may be considered by the driver licensing authority subject to **periodic review** if the standard is met with corrective lenses*. Some discretion is allowed in application of the standard by the treating optometrist or ophthalmologist. However, a driver licence **will not** be issued when visual acuity in the better eye is worse than 6/24. ** Refer to section 10.2.7. Orthokeratology therapy for information on meeting the standard using orthokeratology therapy.*	A person is **not** fit to hold an **unconditional licence**: • if the person's uncorrected visual acuity is worse than 6/9 in the better eye; **or** • if the person's uncorrected visual acuity is worse than 6/18 in either eye. A **conditional licence** may be considered by the driver licensing authority subject to **periodic review** if the standard is met with corrective lenses*. If the person's vision is worse than 6/18 in the worse eye, a **conditional licence** may be considered by the driver licensing authority subject to **periodic review**, provided the visual acuity in the better eye is 6/9 (with or without corrective lenses*) according to the **treating optometrist or ophthalmologist**. The driver licensing authority will take into account: • the nature of the driving task; **and** • the nature of any underlying disorder; **and** • any other restriction advised by the optometrist or ophthalmologist. ** Refer to section 10.2.7. Orthokeratology therapy for information on meeting the standard using orthokeratology therapy.*

Medical standards for licensing – vision and eye disorders		
Condition	**Private standards** (Drivers of cars, light rigid vehicles or motorcycles unless carrying public passengers or requiring a dangerous goods driver licence – refer to definition in **Table 3**)	**Commercial standards** (Drivers of heavy vehicles, public passenger vehicles or requiring a dangerous goods driver licence – refer to definition in **Table 3**)
Visual fields Refer to section 10.2.2. Visual fields.	A person is **not** fit to hold an **unconditional licence**: • if the binocular visual field does not have a horizontal extent of at least 110 degrees within 10 degrees above and below the horizontal midline; **or** • if there is any significant visual field loss (scotoma) within a central radius of 20 degrees of the foveal fixation or other scotoma likely to impede driving performance; **or** • if there is any significant visual field loss (scotoma) with more than four contiguous spots within a 20-degree radius from fixation. A **conditional licence** may be considered by the driver licensing authority subject to **annual review**, taking into account the nature of the driving task and information provided by the **treating optometrist or ophthalmologist**.	A person is **not** fit to hold an **unconditional licence**: • if the person has any visual field defect. A **conditional licence** may be considered by the driver licensing authority subject to **annual review**, taking into account the nature of the driving task and information provided by the **treating optometrist or ophthalmologist** as to whether the following criteria are met: • the binocular visual field has an extent of at least 140 degrees within 10 degrees above and below the horizontal midline; **and** • the person has no significant visual field loss (scotoma, hemianopia, quadrantanopia) that is likely to impede driving performance; **and** • the visual field loss is static and unlikely to progress rapidly.
Monocular vision Refer to section 10.2.2. Visual fields.	A person is **not** fit to hold an **unconditional licence**: • if the person is monocular. A **conditional licence** may be considered by the driver licensing authority subject to **2-yearly review**, taking into account the nature of the driving task and information provided by the **treating optometrist or ophthalmologist** as to whether the following criteria are met: • the visual acuity in the remaining eye is 6/12 or better, with or without correction; **and** • the visual field in the remaining eye has a horizontal extent of at least 110 degrees within 10 degrees above and below the horizontal midline.	A person is **not** fit to hold an **unconditional licence**: • if the person is monocular. A **conditional licence** may be considered by the driver licensing authority subject to **2-yearly review**, taking into account the nature of the driving task and information provided by the **treating optometrist or ophthalmologist**, as to whether the following criteria are met: • the visual acuity in the remaining eye is 6/9 or better, with or without correction; **and** • the visual field in the remaining eye has a horizontal extent of at least 140 degrees within 10 degrees above and below the horizontal midline; **and** • there is no other significant visual field loss that is likely to impede driving performance.

Medical standards for licensing – vision and eye disorders		
Condition	**Private standards** (Drivers of cars, light rigid vehicles or motorcycles unless carrying public passengers or requiring a dangerous goods driver licence – refer to definition in **Table 3**)	**Commercial standards** (Drivers of heavy vehicles, public passenger vehicles or requiring a dangerous goods driver licence – refer to definition in **Table 3**)
Diplopia Refer to section 10.2.3. Diplopia.	A person is **not** fit to hold an **unconditional licence**: • if the person experiences any diplopia (other than physiological diplopia) within 20 degrees from central fixation. A **conditional licence** may be considered by the driver licensing authority subject to **annual review**, taking into account the nature of the driving task and information provided by the **treating optometrist or ophthalmologist** as to whether the following criteria are met: • the condition is managed satisfactorily with corrective lenses or an occluder; **and** • the person meets other criteria as per this section, including visual fields. The following licence condition may apply if corrective lenses or an occluder prevents the occurrence of diplopia: • Corrective lenses or an occluder must be worn while driving. A 3-month non-driving period applies for use of occluders, in order to re-establish depth perception.	A person is **not** fit to hold an **unconditional licence** or a **conditional licence**: • if the person experiences any diplopia (other than physiological diplopia) within 20 degrees from central fixation.

IMPORTANT: The medical standards and management guidelines contained in this chapter should be read in conjunction with the general information contained in Part A of this publication. Practitioners should give consideration to the following:

Licensing responsibility

The responsibility for issuing, renewing, suspending or cancelling a person's driver licence (including a conditional licence) lies ultimately with the driver licensing authority.

Licensing decisions are based on a full consideration of relevant factors relating to health and driving performance.

Conditional licences

For a conditional licence to be issued, the health professional must provide to the driver licensing authority details of the medical criteria not met, evidence of the medical criteria met, as well as the proposed conditions and monitoring requirements.

The presence of other medical conditions

While a person may meet individual disease criteria, concurrent medical conditions may combine to affect fitness to drive – for example, hearing, visual or cognitive impairment (refer to Part A section 2.2.7. Older drivers and age-related changes and section 2.2.8. Multiple medical conditions).

The nature of the driving task

The driver licensing authority will take into consideration the nature of the driving task as well as the medical condition, particularly when granting a conditional licence. For example, the licence status of a farmer requiring a commercial vehicle licence for the occasional use of a heavy vehicle may be quite different from that of an interstate multiple combination vehicle driver. The examining health professional should bear this in mind when examining a person and when providing advice to the driver licensing authority.

Reporting responsibilities

Patients should be made aware of the effects of their condition on driving and should be advised of their legal obligation to notify the driver licensing authority where driving is likely to be affected. The health professional may themselves advise the driver licensing authority as the situation requires (refer to section 3.3 and step 6).